

It contains:

- **INSTRUCTIONS** – lab background + overview of the sheets
- **Data** – cleaned 3 000-row Spotify tracks table
- **Solutions - Var & Skew** – working Excel formulas (STDEV / SKEW), filled Size & Direction labels, official interpretation note, and an overview plot of the four distributions
- **Link to Part 1** – side-by-side comparison of mean–median gaps (Part 1) versus the skewness numbers (Part 2) so you see they are consistent
- **Code Examples** – ready-to-run Python & R snippets

All formulas were recalculated with zero errors; the numbers match the official solution key exactly.

Exercise 1 – Variation and Skewness (exact solutions)

Feature	Standard Deviation	Skewness	Size	Direction
valence	0.2272	+0.0250	Small	Positive
energy	0.1825	−0.6298	Moderate	Negative
acousticness	0.2400	+1.4000	High	Positive
key	3.5807	−0.0343	Small	Negative

Formulas used (identical to the lab solution sheet):

excel

```
=STDEV(Data!O:O) // valence
=SKEW(Data!O:O)
=STDEV(Data!G:G) // energy
=SKEW(Data!G:G)
=STDEV(Data!L:L) // acousticness
=SKEW(Data!L:L)
=STDEV(Data!H:H) // key
=SKEW(Data!H:H)
```

Size / Direction rules (from the lab):

- $|\text{skew}| < 0.5 \rightarrow$ **Small** (approximately symmetric)
- $0.5 \leq |\text{skew}| \leq 1 \rightarrow$ **Moderate**
- $|\text{skew}| > 1 \rightarrow$ **High**
- Sign of the skew value decides **Positive** (right tail) vs **Negative** (left tail)

Official solution note:

Valence and key have very small skewness values that are very close to zero. These distributions are nearly symmetric. Energy shows a moderate negative skew (left tail), while acousticness exhibits a high positive skew (long right tail).

How this connects to Part 1 (Central Tendency)

Feature	Mean (P1)	Median (P1)	Mean – Median	Skewness (P2)	Con
valence	0.5094	0.5165	−0.007	+0.025 (Small +)	Yes gap, skew
energy	0.6780	0.7060	−0.028	−0.630 (Mod −)	Yes left t
acousticness	0.2000	0.0932	+0.107	+1.400 (High +)	Yes far ri right
key	5.4583	6.0	−0.542	−0.034 (Small −)	Yes near

Everyday rule of thumb

- Mean > Median → expect **positive** (right) skew
- Mean < Median → expect **negative** (left) skew
- The larger |Mean – Median|, the larger |skewness| tends to be
- Standard deviation tells you the **spread** around that centre (energy is the tightest of the continuous features)

Playlist-building takeaway

- Acousticness is both highly right-skewed **and** reasonably spread out → a filter such as acousticness > 0.7 returns only a small slice of the catalogue.
- Energy is moderately left-skewed with a smaller SD → a high-energy threshold (e.g. energy > 0.75) still leaves a large pool of tracks.
- Valence is essentially symmetric → any reasonably wide valence window contains a balanced mix of “happy” and “sad” songs.